

Coding & Solution

Date	04 july 2026
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the implementation quality of the Human Development Index (HDI) Prediction Using Machine Learning project. The solution is developed using Python, Flask, and Scikit-learn, following clean coding practices, modular design, and proper documentation. The application successfully predicts the HDI score based on user inputs and provides an easy-to-use web interface.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Field	Details
Repository Link / URL	<i>(Paste your GitHub Repository Link Here)</i>
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Key Features Implemented	HDI prediction using Linear Regression, data preprocessing, model training, Flask web application, user-friendly interface, model serialization using model.pkl
Pending / Incomplete Features	Cloud deployment, user authentication, prediction history, advanced ML models
Setup / Run Instructions	Install dependencies using <code>pip install -r requirements.txt</code> , run <code>python app.py</code> , and open <code>http://127.0.0.1:5000</code> in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/classes	Yes
2	Meaningful variable and	Yes
3	Code includes comments/docum	Yes
4	Error handling is implemented for	Yes
5	The application runs without	Yes
6	Code is committed to a	<i>Yes (if uploaded to GitHub; otherwise change to "No")</i>

Additional Notes / Comments

The project follows a modular folder structure for easy maintenance.

Machine learning and web application logic are separated into different files.

The trained model is saved as model.pkl for efficient prediction without retraining.

The application uses Flask to provide a simple and responsive web interface.

The code is readable, reusable, and can be extended with advanced algorithms such as Random Forest or XGBoost in future versions.